

SEATTLE TACOMA, WA, USA

WMO: 727930

Lat: 47.444N

Lon: 122.314W

Elev: 113

StdP: 99.98

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24233

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.0	-1.1	-12.0	1.4	0.4	-8.6	1.8	1.4	11.9	7.9	10.7	8.4	3.6	10	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.0	18.4	27.9	17.7	26.0	17.0	19.3	28.1	18.4	26.4	17.6	24.7	4.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.6	20.8	15.3	11.0	20.3	14.6	10.5	19.7	54.9	28.3	52.1	26.2	49.6	24.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.2	DB	-5.7	33.7	2.4	2.3	-7.4	35.4	-8.7	36.7	-10.1	38.0	-11.8	39.7	(4)
(5)			WB	-6.9	21.2	2.4	1.8	-8.6	22.5	-10.1	23.5	-11.4	24.5	-13.2	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	5.7	6.2	8.0	10.2	13.6	16.2	19.2	19.2	16.5	11.6	7.8	5.2
	DBStd	5.76	3.22	2.81	2.73	2.78	3.10	3.00	2.87	2.49	2.59	2.43	3.08	3.07
	HDD10.0	593	135	109	75	29	4	0	0	0	0	12	78	151
	HDD18.3	2567	391	341	322	244	151	79	22	18	66	208	317	408
	CDD10.0	1193	2	2	12	35	115	186	284	286	196	62	11	2
	CDD18.3	126	0	0	0	1	4	15	48	46	12	0	0	0
	CDH23.3	1137	0	0	1	8	54	157	453	365	98	1	0	0
	CDH26.7	323	0	0	0	1	10	48	148	99	18	0	0	0
Wind	WSAvg	3.4	3.6	3.7	3.8	3.5	3.3	3.3	3.1	2.9	3.0	3.2	3.6	3.7
Precipitation	PrecAvg	973	141	98	99	70	46	37	17	27	43	91	157	146
	PrecMax	1315	296	231	240	166	94	97	61	117	157	255	397	301
	PrecMin	638	15	9	14	14	3	4	0	0	0	8	19	35
	PrecStd	159	57	51	45	35	24	22	13	29	36	53	70	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	14.8	20.0	24.4	28.0	31.2	33.6	32.1	28.9	21.2	16.5	13.6
		MCWB	10.4	8.9	11.4	14.2	16.4	18.1	19.9	19.1	17.1	14.0	12.8	11.2
	2%	DB	12.2	12.9	16.2	20.0	24.6	27.7	30.5	29.3	26.2	18.6	14.4	11.7
		MCWB	9.4	9.0	10.3	12.3	15.5	17.2	18.9	18.4	16.9	13.3	11.9	9.4
	5%	DB	11.0	11.4	13.9	17.2	21.9	24.6	27.9	27.3	23.8	16.9	13.0	10.4
		MCWB	8.7	8.3	9.5	11.1	14.2	15.9	18.0	17.7	16.2	12.9	11.0	8.3
	10%	DB	10.0	10.2	12.2	15.0	19.3	22.1	25.7	25.1	21.7	15.4	11.9	9.4
		MCWB	7.9	7.5	8.8	10.0	13.3	15.0	17.1	17.2	15.3	12.3	9.8	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	11.7	12.7	15.2	17.5	19.5	21.2	20.1	19.0	16.1	14.2	12.0
		MCDB	13.0	12.6	16.6	22.6	24.5	29.4	31.6	29.2	26.0	18.3	15.5	13.3
	2%	WB	10.5	10.2	11.3	13.0	16.2	17.7	19.6	19.0	17.7	14.7	12.5	10.1
		MCDB	11.4	11.7	14.4	18.1	23.0	25.8	28.7	27.6	24.1	17.0	13.8	11.2
	5%	WB	9.2	9.0	10.4	11.8	15.0	16.5	18.5	18.3	16.8	13.6	11.4	8.8
		MCDB	10.3	10.6	13.2	15.9	21.0	23.3	26.8	26.1	22.4	15.8	12.7	10.1
	10%	WB	8.2	8.0	9.2	10.7	13.8	15.4	17.5	17.5	15.9	12.7	10.3	7.7
		MCDB	9.5	9.7	11.9	14.1	18.6	21.2	24.9	24.4	20.5	14.8	11.6	9.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.9	6.9	7.9	8.7	9.0	10.6	10.3	8.8	6.5	5.4	4.6
		MCDBR	5.8	7.6	9.7	12.0	13.5	13.5	14.5	13.9	12.7	9.1	6.3	5.2
		MCWBR	4.1	4.7	5.2	5.9	6.2	5.7	5.6	5.3	5.3	4.9	4.3	3.8
	5% WB	MCDBR	4.8	5.9	8.2	10.2	11.8	12.1	13.5	12.7	11.2	7.4	5.4	4.7
		MCWBR	4.1	4.3	5.0	5.3	5.9	5.5	5.6	5.3	5.0	4.6	4.2	4.1
Clear-Sky Solar Irradiance	taub	0.324	0.315	0.323	0.355	0.373	0.363	0.349	0.355	0.343	0.327	0.319	0.329	
	taud	2.494	2.548	2.546	2.426	2.396	2.447	2.507	2.497	2.521	2.571	2.542	2.409	
	Ebn at Noon	761	861	906	900	891	900	909	890	870	833	764	695	
	Edh at Noon	65	74	86	107	114	109	101	98	88	72	61	63	
All-Sky Solar Radiation	RadAvg	1.02	1.90	2.85	4.18	5.26	5.79	6.47	5.52	4.02	2.28	1.20	0.83	
	RadStd	0.15	0.25	0.33	0.39	0.43	0.56	0.48	0.37	0.36	0.25	0.16	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.02	N/A	N/A	N/A	N/A	+124	+43
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page